

Worst-Case Bounds: What the Data *Alone* Can Tell You

Supplement to “Mitigation #5: Worst Case Bounds” — Data and Decisions

Basic Idea:

When some people don’t respond, your data are silent about them — but arithmetic still limits what the truth *could* be. Worst-case (or “Manski”) bounds fill in the missing group with the most extreme possibilities and report the resulting range, rather than assuming the missing look like the observed.

Formally

Let r be the response rate, $\bar{y}_R$ the average outcome among respondents, and $\bar{y}_N$ the (unobserved) average among nonrespondents. Then for any outcome that lives between $y_{\min}$ and $y_{\max}$:

$$\text{truth} = r \bar{y}_R + (1 - r) \bar{y}_N \in [r \bar{y}_R + (1 - r) y_{\min}, r \bar{y}_R + (1 - r) y_{\max}].$$

The width of this interval is $(1 - r)(y_{\max} - y_{\min})$: **your ignorance is exactly proportional to the nonresponse rate**, no matter how large the sample.

An Example: Polling

We contact 1,250 voters; 1,000 respond ($r = 0.8$); 60% of respondents support candidate A.

$$\text{Support} \in [0.8 \times 60\% + 0.2 \times 0\%, 0.8 \times 60\% + 0.2 \times 100\%] = [48\%, 68\%].$$

The familiar “ ± 3 ” margin of error is correct *only if* nonrespondents vote exactly like respondents — an assumption (that’s probably wrong). The bounding exercise give us an honest, assumption-free answer is ± 10 around 58%. Note that more data doesn’t help here: contacting 12.5 million voters at the same response rate... the bounds are *still* [48%, 68%]. Sampling error can be reduced with larger sample sizes (money); nonresponse is a problem that sample size can’t fix. At a realistic modern poll response rate of 2.5%, the bounds become [1.5%, 99%] — everything in the headline number is coming from the pollster’s modeling assumptions, not the raw data.

Buying precision with assumptions — explicitly

Bounds turn a hidden assumption into a visible dial. Not willing to assume nonrespondents are identical to respondents, but willing to assume their support lies between 40% and 80%? Then $\text{support} \in 0.8 \times 60\% + 0.2 \times [40\%, 80\%] = [56\%, 64\%]$. Manski calls the general principle the *law of decreasing credibility*: stronger assumptions yield sharper conclusions that fewer people will believe. A good analyst will be clear about the assumptions.

Where you might see this in business

- **Customer satisfaction / NPS.** A 20% survey response rate means an “NPS of 62” has enormous worst-case bounds. Angry and delighted customers respond at different rates — the direction of bias is the interesting question.
- **Employee engagement.** “85% of employees are engaged” from a 60% response rate is consistent with as little as 51% ($0.6 \times 85\%$). Disengaged employees are exactly the ones who skip the survey.
- **Churn and win/loss analysis.** You interview the customers who agree to exit interviews — the ones most likely to leave quietly are missing from the data.
- **Credit & hiring (“reject inference”).** Default rates are observed only for approved applicants; performance only for hired candidates. Bounding is the honest first step before modeling the unobserved group.
- **Clinical trials & program evaluation.** Regulators and top journals increasingly ask for exactly these bounds as sensitivity analyses for patient dropout.

Postmortem

There is something appealing about bounds - assumption free and transparent. Often times they can be so large as to be uninformative (which is itself potentially useful information, I suppose).

Further reading

Manski, C. (1995), *Identification Problems in the Social Sciences*, Harvard Univ. Press — readable treatment (see also his 1989 *J. Human Resources* paper, “Anatomy of the Selection Problem”). Imbens, G. and C. Manski (2004), “Confidence Intervals for Partially Identified Parameters,” *Econometrica* — how sampling uncertainty layers on top of bounds. Meng, X.-L. (2018), “Statistical Paradises and Paradoxes in Big Data,” *Annals of Applied Statistics* — why a huge, non-random sample can be worse than a small random one (e.g., political polling issues, formalized).